InventoryDoc Agent

[illegible]

[illegible] AMD Radeon GPU + ROCm [illegible] AI Agent

[illegible] — [illegible] AI Agent [illegible]

[illegible]

GitHub[illegible]cyslmsolomon

[illegible]1026858211@qq.com

AMD AI DevMaster [illegible] 2026

1. [illegible]

InventoryDoc Agent [illegible] AMD Radeon GPU + ROCm [illegible] AI Agent[illegible] 10 [illegible]10 [illegible] 98.99 [illegible]

1.1 [illegible]

[illegible]

- [illegible]10+ [illegible]CSV/XLSX/TXT[illegible]
- [illegible]NOISE [illegible]
- [illegible] 2-3 [illegible]
- [illegible]

1.2 [illegible]

- [illegible]
- [illegible]
- [illegible]

2. [illegible]

2.1 [illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible] 3 [illegible]NOISE [illegible]	[illegible]
[illegible]	[illegible]/[illegible]/[illegible]	[illegible]
[illegible]	[illegible]/[illegible]/[illegible]	[illegible]

2.2 [illegible]

- **[illegible]RAG[illegible]**— [illegible]
- [illegible] — Excel [illegible] 10+ [illegible]
- [illegible] — 10 [illegible]
- [illegible] — [illegible]
- [illegible] — 7 [illegible] must_check_points [illegible]

3. 10 [illegible]

[illegible]	[illegible]	[illegible]
Step 1	[illegible]	[illegible] CSV/XLSX/TXT[illegible]
Step 2	[illegible]	NOISE/[illegible]/[illegible]
Step 3	[illegible]	[illegible]/[illegible]/[illegible]
Step 4	[illegible]	NOISE [illegible] + [illegible]
Step 5	[illegible]	[illegible]
Step 6	[illegible]	[illegible] + [illegible]
Step 7	[illegible]	[illegible]/[illegible]/[illegible]
Step 8	[illegible]	[illegible]/[illegible]/[illegible]
Step 9	[illegible]	[illegible] + [illegible]
Step 10	[illegible]	Excel 4 sheets + MD [illegible]

[illegible]7 [illegible] must_check_points [illegible]

4. [illegible]

4.1 [illegible]

[illegible]

- [illegible]Gradio Web UI[illegible] 4 [illegible]
- **Agent** [illegible]LangChain + 10 [illegible]
- **RAG** [illegible]
- [illegible]Excel [illegible]
- [illegible]vLLM + ROCm[illegible]AMD GPU [illegible]

4.2 [illegible]

[illegible]	[illegible]	[illegible]
Agent Core	src/agent.py	10 [illegible]
RAG Engine	src/rag.py	[illegible]
Noise Detector	src/noise_detector.py	[illegible]
Tools	src/tools.py	Excel [illegible]/[illegible]/[illegible]
Memory	src/memory.py	[illegible]
Quality Checker	src/quality_checker.py	7 [illegible]
UI	src/ui.py	Gradio Web [illegible]

5. AMD GPU □□□□□

5.1 □□□□

□□□□□ AMD Radeon GPU + ROCm □□□□□□□□□□□□□□□□ AI Agent □□□□□□□□□□□□ AMD GPU □□□□□□□□□□□ API□

□□	□□□□	□□
GPU □□	AMD Radeon RX 7900 XTX	24GB VRAM□□□ FP16/BF16
□□□□	vLLM 0.6+	□□□ LLM □□□□
□□□□□□	PyTorch 2.4+ ROCm	AMD GPU □□□□
□□□□	BAAI/bge-small-en-v1.5	□□□□□□GPU □□
□□□□□	ChromaDB	□□□□□□

5.2 □□□□□□

□□□□	□□	□□
FP16 □□□□	□□ float16 □□□□	□□□□□□ 50%
Continuous Batching	vLLM □□□□□	□□□□□ 2-4 □
PagedAttention	vLLM □□□□	□□□□ batch size
GPU □□□□□□	□□□□□ GPU □	□□□□□□ 60%

5.3 □□□□□ AMD ROCm

□□	**AMD ROCm**	**NVIDIA CUDA**
□□□	□□□□	□□
□□	□□□□□□	□□ GPU □□
□□□□	□□□□□□	□□ NVIDIA
□□□□□	vLLM □□□□	vLLM □□□□

6. [illegible]

6.1 Ground Truth [illegible]

[illegible]

[illegible]	Agent [illegible]	[illegible]	[illegible]
SKU [illegible]	528	528	[illegible]
[illegible]	21,144 [illegible]	21,144 [illegible]	[illegible]
[illegible]	-971 [illegible]	-971 [illegible]	[illegible]
[illegible]	4.6%	4.6%	[illegible]
[illegible]	-989,900.97 [illegible]	-989,900.97 [illegible]	[illegible]
[illegible]	14/21/124	14/21/124	[illegible]
[illegible]	7/7 [illegible]	[illegible]	[illegible]

6.2 [illegible]

[illegible]	[illegible]	Agent	[illegible]
[illegible]	2-3 [illegible]	10 [illegible]	99%
[illegible]	[illegible]	100%	—
[illegible]	[illegible]	7 [illegible]	—

7. [illegible]

[illegible]	[illegible]
[illegible]	[illegible]/[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

8. [illegible]

- [illegible]PDF [illegible]
- [illegible]
- [illegible]
- [illegible] ERP [illegible]

9. [illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]Agent [illegible]

10. [illegible]

- GitHub[illegible]https://github.com/cyslmsolomon
- [illegible]1026858211@qq.com
- PR [illegible]https://github.com/AMD-DEV-CONTEST/Radeon-hackathon-2026-07/pull/26